Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton – Flowering and Fruiting

1.1 Floral Initiation, Structure, and Flowering Pattern

- **Flowering stages and development:** Cotton (*Gossypium* spp.) shifts from vegetative growth to reproductive development when axillary or terminal meristems differentiate into reproductive branch primordia. The apical meristematic dome produces three modified triangular leaf-like organs termed epicalyx bracts that enclose the meristematic dome and terminate its main-stem elongation, forming a flower bud or "square". The square becomes visually recognizable as a "pinhead square" approximately 21 days prior to bloom when the subtending leaf unfolds. On the day of anthesis, a white to pale-yellow flower opens at dawn, turns pink-to-red on the second day post-anthesis, and dries into a dark reddish-brown bloom tag that caps the expanding green boll.

- **Flower morphology and reproductive biology:** Cotton flowers are complete, hermaphroditic, and actinomorphic. The floral architecture consists of:

  1. *Epicalyx:* Three large, triangular, laciniate or entire leaf-like bracts that enclose and protect the inner floral organs.
  2. *Calyx:* A green, cup-shaped structure formed by five fused sepals.
  3. *Corolla:* Five large, overlapping petals ranging from creamy-white to yellow, frequently displaying a deep reddish-purple basal spot in *Gossypium arboreum* (Desi cotton).
  4. *Androecium:* A tubular stamen column (monadelphous structure) bearing numerous functional, yellow anthers.
  5. *Gynoecium:* A central pistil comprising a superior 3-to-5 carpellate ovary containing multiple ovules in axile placentation, an elongated style, and a receptive, lobed stigma.

  Under environmental stress, plants can form four-bract squares; these fail to interlock properly, leaving structural gaps that allow pest vectors and bacteria to enter, predisposing young squares to bacterial soft rot and abortion.

- **Flowering pattern and spatial distribution:** Flowering follows a structured, predictable spatial timeline across the canopy:

  – *Vertical Flowering Interval (VFI):* A flower opens approximately every 2.5 to 3.0 days at successive vertical main-stem nodes.

  – *Horizontal Flowering Interval (HFI):* Adjacent flowers along the same sympodial fruiting branch open at intervals of 3.8 to 6.0 days.

Up to 80% of the final lint yield is concentrated in first- and second-position bolls located on main-stem nodes 9 through 14 due to higher photosynthetic assimilate supply from nearby subtending and main-stem leaves.

- **Branching architecture and node positioning:** Cotton exhibits an indeterminate growth habit featuring two distinct branch types:
  - *Monopodial Branches (Vegetative Limbs):* Arise from lower main-stem nodes (nodes 1–4) and grow continuously from a single apical meristem without directly terminating in a floral structure.
  - *Sympodial Branches (Fruiting Branches):* Arise from higher main-stem nodes as cymose inflorescences. Each sympodial unit produces a node, internode, subtending leaf, and a terminal square, followed by secondary axillary elongation that creates a characteristic "zigzag" branch architecture.

  The main-stem node position where the first sympodial branch develops is termed the Node of First Fruiting Branch (NFFB), which serves as a primary morphological index for crop earliness.

1.2 Environmental Requirements and Phenological Timelines

- **Environmental requirements:** Cotton flowering and fruiting require warm conditions with an optimal daytime temperature range of 25–32°C and night temperatures between 18–22°C. High solar radiation and moderate ambient humidity favor photosynthetic assimilate production and pollen tube elongation, whereas prolonged cloudiness, excessive rainfall during anthesis, or drought severely impair pollination, flower retention, and boll setting.
- **Flowering time and phenological duration:** Crop phenology in the major growing zones (e.g., North, Central, and South India) adheres to strict seasonal timelines:
  - *Sowing Window:* Completed between mid-April and May in northern canal-irrigated zones, or June–July in rainfed central/southern zones.
  - *Squaring Stage (Pre-flowering):* Occurs between 45 and 70 Days After Sowing (DAS).
  - *Peak Flowering and Boll Setting:* Spans 70 to 120 DAS.
  - *Boll Development and Maturity:* Spans 120 to 150 DAS.
  - *Harvesting / Boll Bursting:* Spans 150 to 180+ DAS. Total crop duration ranges from 150 days in early-maturing cultivars to 180–190 days in long-duration hybrids.

1.3 Pollination, Fertilization, and Fiber/Boll Development

- **Pollination and fertilization:** Cotton is primarily a self-pollinated crop. Anther dehiscence occurs on the morning of anthesis between 0700 and 1100 h, shedding pollen grains directly onto the receptive stigma. Pollen grains germinate within 30 minutes, extending pollen tubes down the style toward the ovules. Double

fertilization is completed within 12 to 24 hours post-anthesis, forming the diploid zygote and triploid endosperm.

- **Boll setting and initial development:** Following double fertilization, the superior ovary transitions into rapid cell division and expansion, giving rise to the green boll. Fiber and seed development proceed concurrently across four distinct physiological stages:

  1. *Fiber Initiation (0 to 5 Days Post-Anthesis, DPA):* Single epidermal cells on the ovule outer integument differentiate into initial protuberances (16,000 initials/ovule) driven by vacuolar turgor and sucrose synthase activity.
  2. *Fiber Elongation and Early Embryogenesis (2 to 20 DPA):* Single-celled fibers undergo extreme polar expansion (reaching 2–3 cm), held in organized bundles by an adhesive Cotton Fiber Middle Lamella (CFML). The embryo progresses from globular to heart and torpedo stages.
  3. *Secondary Wall Synthesis (16 to 40 DPA):* Enzymatic breakdown releases individual fibers, followed by concentric inner deposition of pure cellulose (>90%), establishing final fiber strength and maturity. Cotyledons absorb the endosperm by 24 DPA and accumulate storage lipids and proteins.
  4. *Maturation and Desiccation (45 to 50 DPA):* Vascular connections in the funiculus dry up, the seed and fiber desiccate, and the boll cracks open. Fibers collapse into flattened, convoluted ribbons suitable for textile spinning.

1.4 Environmental Stresses and Reproductive Failure

- **Environmental stresses and reproductive failure:** Extremely high daytime temperatures (>35–38°C) or warm night temperatures (>22°C) deactivate Rubisco activase, reduce Photosystem II efficiency, disrupt polyamine synthesis (spermidine and spermine in ovaries), and inhibit pollen germination and tube growth through the style, causing severe flower drop and failure of fertilization.

- **Water-deficit stress:** Water stress imposed prior to flowering inhibits the initiation of fruiting sites. Water deficits during peak flowering and boll setting trigger an immediate spike in ethylene synthesis and abscisic acid (ABA) accumulation in pedicel abscission zones. Ethylene activates hydrolytic enzymes (cellulase and pectinase), inducing cell wall breakdown and causing massive shedding of squares and young bolls (<14 DPA).

- **Waterlogging and oxygen stress:** Saturation of heavy soils or water stagnation for over 24 hours halts root respiration, causes nutrient uptake failure, and triggers sudden wilting or parawilt.

- **Carbohydrate stress and cutout:** Developing bolls represent massive carbohydrate sinks. As boll load accumulates, photoassimilates supplied to the main-stem apical meristem are exhausted. Main-stem elongation and sympodial node production cease, a physiological state defined as Nodes Above White Flower reaching five ($NAWF = 5$), or "cutout".

1.5 Water and Nutrient Management during Flowering

- **Water management:** Peak flowering and boll setting (70–120 DAS) represent the most water-sensitive phase of cotton growth, during which moisture stress must not be permitted. Flood/furrow irrigation is applied at scheduled intervals of 10–12 days during peak bloom. Prolonged dry spells followed by heavy flooding must be avoided to prevent parawilt and boll cracking. All irrigation is terminated by the end of September to encourage natural boll drying and opening.
- **Nutrient management:** Cotton exhibits high Nitrogen (N) and Potassium (K) requirements during flowering. Fertilizer application schedules require:
  - Basal application of full Phosphorus (DAP or SSP), 10 kg/acre Zinc Sulphate (21%), 20 kg/acre Muriate of Potash (MOP), and 400 g/acre Boron (4 kg borax) in deficient soils.
  - Nitrogen is applied in splits: 50% at thinning/early growth and 50% top-dressed at flower initiation. Excessive early N application must be avoided to prevent rank vegetative growth.
- **Foliar nutritional sprays:** To meet high reproductive sink demands and manage leaf reddening in Bt cotton:
  - Apply 3 to 5 foliar sprays of 2% Potassium Nitrate (KNO_3 / 13:0:45) at weekly intervals starting at 10% flower initiation.
  - Spray 1% Magnesium Sulphate ($MgSO_4$ @ 1 kg/100 L water) twice at 15-day intervals during full bloom.
  - Spray 2% DAP or 2% Urea following waterlogging or crop recovery.

1.6 Plant Growth Regulation and Canopy Management

- **Plant growth regulation:** Foliar plant growth regulators are deployed to adjust vegetative-to-reproductive balance:
  - *Mepiquat Chloride (PIX):* Applied @ 300 mL/acre (1.0–1.2 mL/L water) at 60 and 75 DAS. Mepiquat chloride inhibits gibberellic acid biosynthesis, reduces main-stem internodal length, compacts canopy architecture, lowers internal microclimate humidity, and hastens cutout by 4–6 days to prevent crop lodging.
  - *Naphthalene Acetic Acid (NAA / Planofix):* Applied @ 20 ppm (7 mL / 15 L water) twice at 15-day intervals during peak squaring and flowering to inhibit cellulase and pectinase in the abscission zone, significantly suppressing square and boll shedding.
  - *1-Methylcyclopropene (1-MCP):* Ethylene action inhibitor applied at first bloom to suppress stress-induced ethylene peaks and improve middle-canopy boll retention under drought or temperature stress.
- **Canopy and crop management:** Agronomic practices include:

- *Detopping (Terminal Bud Nipping):* Removing the top 2–3 cm of the main-stem terminal bud when plants reach 100–120 cm height (90–100 DAS) halts vertical growth and redirects photoassimilates directly to sympodial boll filling.
- *Monopodial Branch Removal:* Pruning the lowest 2 to 3 vegetative branches eliminates non-fruiting nutrient sinks, improves sunlight penetration and aeration, and reduces microclimate humidity.
- *Border and Intercropping:* Planting 2 rows of maize or sorghum along field borders suppresses sucking pests and protects developing flowers.

1.7 Pests and Diseases Affecting Reproduction

- **Pests affecting reproductive structures:**
  - *Pink Bollworm (Pectinophora gossypiella):* Major threat during flowering (70–120 DAS). Larvae feed on anthers and stigmas inside flowers, webbing petals into a characteristic "rosette flower" appearance, and bore into green bolls to feed on seeds, staining lint and causing internal lock rot.
  - *American Bollworm (Helicoverpa armigera) & Spotted Bollworm (Earias spp.):* Larvae bore into squares, flowers, and green bolls, producing flared-up bracts, large circular entry holes surrounded by frass, and heavy premature dropping.
  - *Mirid Bugs (Creontiades biseratense & Campylomma livida):* Nymphs and adults suck sap from squares and young bolls, causing dark sunken lesions, square shedding, and deformed "parrot-beaked" bolls.
  - *Sucking Pests (Whiteflies, Thrips, Jassids, Aphids):* Extract sap from leaves and floral parts, excrete sticky honeydew that fosters black sooty mould growth, impair photosynthesis, and vector Cotton Leaf Curl Virus (CLCuV) or Tobacco Streak Virus (TSV).
  - *Blister Beetle (Mylabris pustulata) & Flower Chafer Beetle (Oxycetonia versicolor):* Devour petals, anthers, and stigmas during peak flowering, directly preventing pollination and boll set.
- **Diseases affecting reproduction:**
  - *Bacterial Blight (Xanthomonas citri pv. malvacearum):* Causes angular leaf spots, black arm stem lesions, and water-soaked depressed spots on green bolls, staining lint yellow-brown.
  - *Boll Rot Complex:* Fungal rots (*Colletotrichum*, *Fusarium*, *Alternaria*) and bacterial rot (*Pantoea* spp., vectored by stink bugs and thrips) infect green bolls, turning internal locks into a wet, slimy yellow/reddish mass.
  - *Fungal Wilts & Root Rots: Fusarium oxysporum f. sp. vasinfectum* and *Rhizoctonia solani* cause root decay, vascular discoloration, drooping, and sudden drying during heavy boll loading.
  - *Foliar Spotting & Mildews:* Alternaria leaf spot (*A. macrospora*), Target spot (*Corynespora cassiicola*), and Grey mildew (*Ramulariopsis gossypii*) cause severe leaf blighting, defoliation, and premature crop senescence.

1.8 Flower Drop, Shedding, and Physiological Disorders

- **Flower drop and fruit-setting disorders:** Shedding of squares, flowers, and young bolls occurs naturally or is exacerbated by high temperature stress (>35°C), water deficit or waterlogging, low sunlight, nutrient deficiencies, and pest feeding. Ethylene and ABA buildup in the pedicel abscission zone triggers cell wall hydrolysis, leading to detachment of fruiting bodies. Natural fruit shedding frequently reaches 50–60% of total initiated fruiting sites.
- **Parawilt / Sudden Wilt Complex:** A severe physiological disorder occurring when dry, hot periods or waterlogged/saturated soils are followed by bright, intense sunshine. This creates a severe breakdown in the Soil-Plant-Atmosphere Continuum (SPAC), where extreme transpiration demand exceeds root water/nutrient uptake capacity. Affected plants exhibit sudden wilting, leaf drooping, chlorosis, bronzing, and premature shedding of squares and young bolls.
- **Leaf Reddening in Bt Cotton:** Triggered by Nitrogen and Magnesium deficiencies combined with low night temperatures during peak boll load, causing chlorophyll breakdown and red anthocyanin pigment synthesis in leaves.
- **Parrot-beaked bolls:** Asymmetric boll deformity caused by mirid bug or thrips feeding damage on one or two locules of young developing bolls.

1.9 Growth Monitoring, Crop Performance, and Integrated Workflow

- **Fruit-setting assessment and growth monitoring:** Regular field monitoring and Agro-Eco System Analysis (AESA) track:
  - *Nodes Above White Flower (NAWF):* Tracks physiological crop status; $NAWF = 5$ indicates physiological cutout.
  - *Economic Threshold Levels (ETL):* Initiate control measures when bollworm infestation reaches 5% damaged fruiting bodies or 1 larva/plant; Pink Bollworm moth catches reach 8 moths/trap/day for 3 consecutive days; Mirid bugs reach 2–3 bugs/plant; or Whitefly populations reach 5–10 nymphs/adults per leaf.
  - *Trap Monitoring:* Install pheromone traps @ 5 traps/ha and yellow sticky traps @ 25 traps/ha.
- **Effects on crop performance:** Final yield is determined by the total number of harvested bolls per plant, average boll weight (3.5–6.0 g), Ginning Outturn (GOT 32–40%), fiber length (24–32 mm), fiber strength, and micronaire value. Concentrating boll retention on main-stem nodes 9–14 maximizes overall seed cotton production (20–35 q/ha under irrigated conditions).
- **Integrated management workflow:**
  1. *Pre-sowing Phase:* Deep summer ploughing, selecting resistant hybrids/varieties, acid delinting, and seed treatment with Carboxin 37.5% + Thiram 37.5% @ 3.5 g/kg or *Trichoderma* @ 5 g/kg seed.

2. *Vegetative to Squaring Phase (0–60 DAS):* Apply basal NPK + 10 kg $ZnSO_4$; deploy border barrier crops (maize/sorghum); monitor sucking pests using yellow sticky traps; thin crowded seedlings.
3. *Flowering and Boll Setting Phase (70–120 DAS):* Apply second split N dose; spray 2% Potassium Nitrate (KNO_3) 3–5 times at weekly intervals; apply NAA (Planofix) @ 20 ppm to prevent shedding; apply Mepiquat Chloride @ 1.0–1.2 mL/L if rank vegetative growth occurs; detop main stem at 100–120 cm height; spray targeted insecticides/biopesticides for Pink Bollworm, mirids, and boll rot upon reaching ETL.
4. *Boll Maturity and Harvest (120–180 DAS):* Terminate irrigation by late September; allow natural boll cracking; perform clean pickings to prevent lint staining and trash contamination.